Hantao Jiang

Shanghai Jiao Tong University | (+86)19877376168 | eric_jht@sjtu.edu.cn

*A third-year undergraduate interested in **Computer Vision, Machine Learning and Generative models**.*
Concentrating on Training-free Controllers for Vision Generation and Efficient Generative Models.
Sincere desire in deeper research experience and broader learning.

Education & Knowledge

(Current) ACM Honor Class, Shanghai Jiao Tong University, GPA: 3.73/4.0
Mentored by Prof. Xiaohong Liu, SJTU Multimedia Lab

Relevant Courses [Knowledge Field]
Computer Vision (100/100), Online Algorithms, Foundations of Data Science, Machine Learning, Information Theory, Optimization Methods, Probability Theory

Academic Activities & Skill Stack

Course Project on Visuo-Tactile Associate Representation Learning

- Co-designed Mask-Annealing Training Strategy and Mixture of Transformers | deep understanding on diffusion-based generation model structure, project-level finetuning pipeline design experience
- Organized image-mask dataset and experiment codebases | customizing data preprocessing pipeline with Opencv | VLM-assisted quality control | auto data sourcing | weak-supervised preprocessing
- Basic skills attained: PyTorch framework, WandB experiment tracing, docker image deployment

Long-term Internship in SJTU Multimedia Lab
Shanghai | June 2025 – Present

- Insights on the role of tactile sensing for VLAs: touch conditioner for contact-rich manipulations; complementary information for vision occlusion area; auxiliary but cheap enhancement
- Proposed TactiLDM, a professional tactile encoder with multi-stage training strategy | DINO-style contrastive learning | multimodal conditioning
- Reinforcement adaptation approach in simulated manipulation tasks | gymnasium-based physical environment simulation

Ongoing Projects on Controlled Generation & 3D Reconstruction

- Learned mathematical principles: flow matching, forward/backward guidance & multi-view geometry, SfM, NeRF, Gaussian splatting, KinectFusion, Poisson SR……
- In-practice paradigms: training-free style-content decoupling, attention-based semantic guidance
- Credible coding ability derived from reproduction of papers

Honors

- Full-time Zhiyuan Honors Scholarship | Awards for excellence